Microcontroller Based Systems & Robotics Hub

Promoting robotic design and entrepreneurship experiences among students and teachers

Lesson 12:
Advanced Arduino Programming -I

CONTENTS



- Decimal -binary system
- Math operators

- TASK/ACTIVITY:Do it yourself –example problems

DECIMAL TO BINARY CONVERSION

- Decimal system: A numeral system whose numbers are represented with digits 0-9
- Thenumber is expressed as abase-10 reference system
  - o Examples: 2 10,510,1010, 10010, etc.

- Binary system: A numeral system whose numbers are represented with digits 0 and 1 only
- Thenumber is expressed as abase-2 reference system
  - o Examples: 00 2, 102 (10 is not ‘number ten’ but equals to ‘number two’ in the base 2 system), 012, 0102 etc.

DECIMAL TO BINARY CONVERSION

to binary number

- The process of converting a decimal number is called successive division

  Process:

-

1. Divide thedecimal number by 2
2. Theremainder is the Least Significant Bit (LSB) of binary number
3. The division is continued until the quotient is zero, then the conversion is complete of the binary number
4. The new remainder is the next Most Significant Bit (MSB) in every successive division step

DECIMAL TO BINARY CONVERSION

- Example 1: Convert decimal number 6 (base 10) to binary number (base 2)

$$2 \overline{)\,6}\ \ 3 \quad \text{Remainder} = 0 \rightarrow \text{LSB}$$

$$2 \overline{)\,3}\ \ 1 \quad \text{Remainder} = 1 \rightarrow \text{next MSB}$$

$$2 \overline{)\,1}\ \ 0 \quad \text{Remainder} = 1 \rightarrow \text{MSB}$$

$\therefore 6_{10} = 110_2$

DEC → BINARY: ACTIVITY

Problem 1 : Do it yourself!

Convert the decimal number 26_{10} into its binary equivalent

DEC → BINARY: SOLUTION

Problem 1: Convert the decimal number 26_{10} into its binary equivalent

Solution:

$2 \overline{)\,26}$ — 13 — Remainder = 0 → LSB

$2 \overline{)\,13}$ — 6 — Remainder = 1 → next MSB

$2 \overline{)\,6}$ — 3 — Remainder = 0

$2 \overline{)\,3}$ — 1 — Remainder = 1

$2 \overline{)\,1}$ — 0 — Remainder = 1 → MSB

$\therefore 26_{10} = 110102$

DEC → BINARY: ACTIVITY

Problem 2: Do it yourself!

Convert the decimal number 41_{10} into its binary equivalent

DEC → BINARY: SOLUTION

Problem 2: Convert the decimal number 41_{10} into its binary equivalent

Solution:

$2 \overline{)\,41}$ — Quotient = 20 — Remainder = 1 → LSB

$2 \overline{)\,20}$ — Quotient = 10 — Remainder = 0 → next MSB

$2 \overline{)\,10}$ — Quotient = 5 — Remainder = 0

$2 \overline{)\,5}$ — Quotient = 2 — Remainder = 1

$2 \overline{)\,2}$ — Quotient = 1 — Remainder = 0

$2 \overline{)\,1}$ — Quotient = 0 — Remainder = 1 → MSB

$\therefore 41_{10} = 1010012$

DEC → BINARY: ACTIVITY

ACTIVITY

a) $13_{10} = ?$

b) $22_{10} = ?$

c) $43_{10} = ?$

d) $158_{10} = ?$

DEC → BINARY: SOLUTIONS

a) 13_{10} = ? 1101_2

b) 2210= ? 101102

c) 4310= ? 101011_2

d) 15810= ? 100111102

BINARY TO DECIMAL CONVERSION

- The process of converting a binary number to decimal number is called weightedmultiplication

- Process:
  1. The decimal number will be equal to the sum of each binary digit starting from the left-most digit times their power of 2 (2n), n = 0, 1, 2, .., (n – 1) which is called the bit-weighting factor, where n is the no. of digits in thebinarynumber
  2. Example: Bit 0 → 20 = 1, Bit 1 → 21 = 2, Bit 2 → 22 = 4, etc.

BINARY TO DECIMAL CONVERSION

Example:

Convert the decimal number 01102into its decimal equivalent where n = 4

Bit 4		Bit 3		Bit 2		Bit 1		
0		1		1		0		
x		x		x		x		
23 = 8		22 = 4		21 = 2		20 = 1		
‖		‖		‖		‖		
0	+	4	+	2	+	0	=	610

→ n

→ Binary digits

→ Bit-weighting Factors

→ Corresponding decimal digits

∴01102= 6 10

BINARY → DEC: ACTIVITY

Problem 1: Do it yourself!

Convert the binary number 100102into its decimal equivalent where n = 6

BINARY → DEC: SOLUTION

Problem 1: Convert the binary number 10010 into its decimal equivalent where n = 5

Solution:

Bit 5		Bit 4		Bit 3		Bit 2		Bit 1		
1		0		0		1		0		
X		X		X		X		X		
$24 = 16$		$23 = 8$		$22 = 4$		$21 = 2$		$20 = 1$		
||		||		||		||		
16	+	0	+	0	+	2	+	0	=	18_{10}

$\therefore 100102 = 18_{10}$

BINARY → DEC: ACTIVITY

a) 0110 2 = ?

b) 11010 2 = ?

c) 0110101_2 = ?

d) 110100112= ?

BINARY → DECIMAL : SOLUTIONS

a) 0110 2 = ? 6 10

b) 11010_2 = ? 26_{10}

c) 01101012 = ? 53_{10}

d) 110100112= ? 211_{10}

DECIMAL AND BINARY –REVIEW

Base $_{10}$ DECIMAL → Successive Division → Base2 BINARY

a) Divide the decimal number by 2, remainder = LSB of binary number

b) If the quotient = zero, the conversion is complete; else repeat step (a) using the quotient = decimal number and new remainder = next MSBof the binary number

Base $_{2}$ BINARY → Weighted M ultiplication → Base $_{10}$ DECIMAL

a) Multiply each bit of the binary number by it corresponding bit-weighting factor (i.e., Bit -0 → 20 = 1, Bit –1 → 21 = 2, etc.)

b) Sumup all the products in step (a) to get the decimal number

MATH OPERATORS

- abs(x): absolute value of x (applicable if x = int and long)
- fabs(x): absolute value of x (applicable if x = float)
- min(x, y) and max(x, y): minimum and maximum of two arguments x and y
- pow(x, y): x ^y
- sq(x): x ^2
- Trigonometric functions: sin(x), cos(x), x has to be in radians
- random(max): generates a random number between 0 and max
- random(min, max): generates a random number between min and max

MATH OPERATOR

ACTIVITY

1.Write a program to display minimum of the absolute values of 40 and -30

1.Write a program to find the value of sin(x) + (cos(x))2at x = 1.5 radians

MATH OPERATOR –SOLUTION FOR 1st

Minimum_value | Arduino 1.8.19

File Edit Sketch Tools Help

Minimum_value §

```
int x = 30;
int y = -40;
void setup() {
  Serial.begin(9600);

  int z = min(x, y);

  Serial.print("x    = ");
  Serial.println(x);
  Serial.print("y    = ");
  Serial.println(y);
  Serial.print("Minimum value of 40 and -30 = ");
  Serial.println(z);
}

void loop() {}
```

OUTPUT:

The minimum value of the array is: -40

MATH OPERATOR –SOLUTION FOR 2nd

Equation_with_angles | Arduino 1.8.19

File Edit Sketch Tools Help

Equation_with_angles §

```
int angle=1.5; // in radians
int val;

void setup() {
// put your setup code here, to run once:
Serial.begin(9600);
val = sin(angle) + sq(cos(angle));
Serial.print("The value of the equation at 1.5 radians is: ");
Serial.println(val);
}

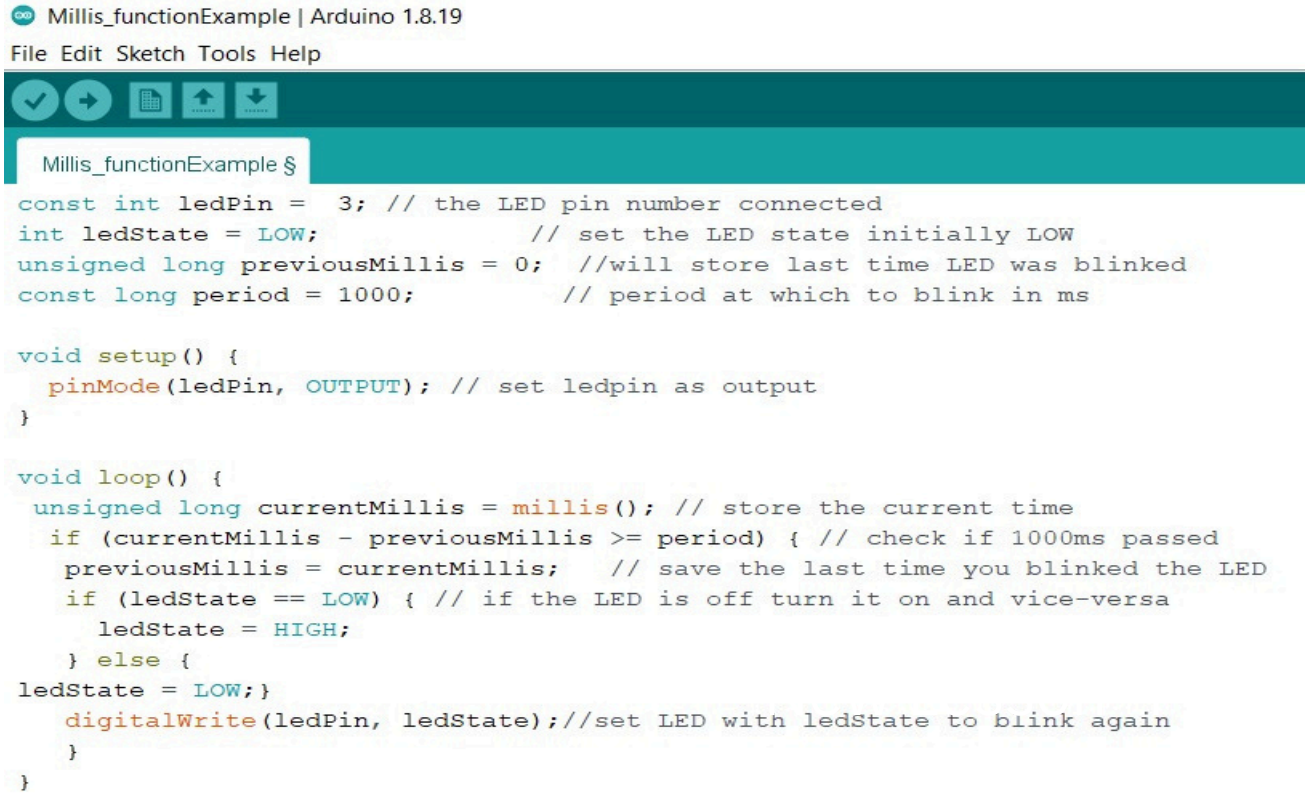
void loop() {}
```

OUTPUT:
The value of the equation at 1.5 radians is: 1

PROGRAMMING: OTHERS

delay()	delay in milliseconds
delayMicroseconds	delay in microseconds
millis()	Time in milliseconds since the Arduino board began running the current program
micros()	Time in microseconds since the ArNdueinwo bLoaersd sboeg a current program

MILLIS FUNCTION EXAPMLE




```
const int ledPin =  3; // the LED pin number connected
int ledState = LOW;             // set the LED state initially LOW
unsigned long previousMillis = 0;  //will store last time LED was blinked
const long period = 1000;          // period at which to blink in ms

void setup() {
  pinMode(ledPin, OUTPUT); // set ledpin as output
}

void loop() {
 unsigned long currentMillis = millis(); // store the current time
  if (currentMillis - previousMillis >= period) { // check if 1000ms passed
   previousMillis = currentMillis;   // save the last time you blinked the LED
   if (ledState == LOW) { // if the LED is off turn it on and vice-versa
     ledState = HIGH;
   } else {
ledState = LOW;}
   digitalWrite(ledPin, ledState);//set LED with ledState to blink again
   }
}
```

Thank You!

Questions and Feedback?